

Proof that $\sqrt{2}$ is Irrational

Assume, for contradiction,

$$\sqrt{2} = \frac{p}{q} \left| \begin{array}{l} p, q \in \mathbb{Z}^+ \wedge \gcd(p, q) = 1. \end{array} \right.$$

$$\sqrt{2} = \frac{p}{q}$$

$$2 = \frac{p^2}{q^2}$$

$$2q^2 = p^2.$$

$$2 \mid p^2 \implies 2 \mid p.$$

Let

$$p = 2k \quad k \in \mathbb{Z}^+.$$

$$2q^2 = (2k)^2$$

$$2q^2 = 4k^2$$

$$q^2 = 2k^2.$$

$$2 \mid q^2 \implies 2 \mid q.$$

Thus

$$2 \mid p \wedge 2 \mid q \implies \gcd(p, q) \neq 1,$$

contradicting $\gcd(p, q) = 1$.

$$\therefore \sqrt{2} \notin \mathbb{Q}.$$

Hence, $\sqrt{2}$ is irrational. ■